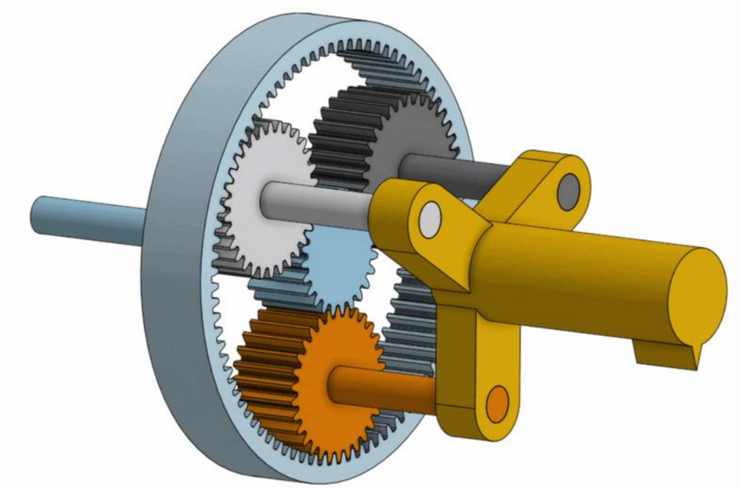
CAD PROJECTS:

(excluding IP protected TJO and Treace work)

Planetary Gear:



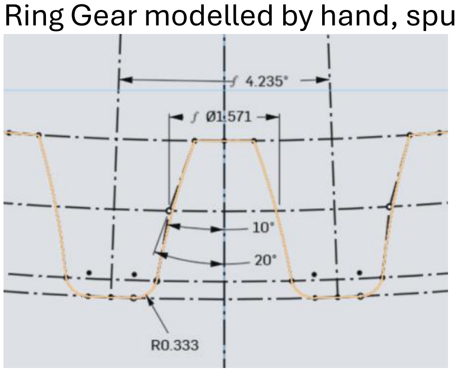
SUN: 24 teeth= N_S
PLANET: 30 teeth= N_P
RING: 84 teeth= N_R
Teeth of ring and sun need to be integers divisible by 3.

Gear Ratios:

Orbit Ratio between the Planet gear rotating around the Ring can be modelled by $\frac{N_P}{N_R} = \frac{30}{84} = 0.357$ or for every 1 rotation of a planet gear, the planet makes 0.357 of an orbit around the ring.

The turn ratio between the sun gear and the planet carrier is $\frac{N_S}{N_S+N_R} = \frac{24}{24+84} = 0.222$

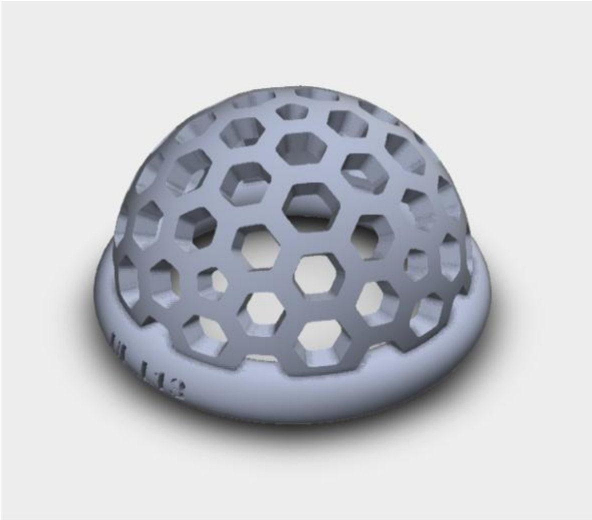
Ring Gear modelled by hand, spur gears made from an onshape add-on



Teeth for all gears have a module of 1mm, a pitch of pi, a pressure angle $\alpha = 20°$and a root fillet of 1/3. Angle between repeating parts for the ring gear should be $\frac{1}{85} * 360 = 4.235°$

Sensor Rig for a compound eye camera

Hex lattice patterned across a hemisphere using SolidWorks revolved boss-extrude with circular pattern features. Inspired by compound eyes and icosphere geometry in Blender.



Senior Design Project, a mechanical nerve tensiometer

Assembly includes 7custom parts and 2 off-the-shelf components, with sliding mechanism designed to translate axial tension into a measurable readout